

CS504-Software Engineering-I

Solved MCQS for Final terms papers

Solved by JUNAID MALIK and Team



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- 2) *DD*
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- 4) *Viva preparation*
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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. Thin Client Model is one of the configurations of _____ Model.
 - a. Data-centered
 - b. Client Server** page130
 - c. Layered
 - d. Reference
2. STL Stands for _____
 - a. Standard Template Library** page 138
 - b. Standard Type Link
 - c. Standard Tempo Line
 - d. Standard Type Link
3. The three basic principles that guide maintainability are:
 - (a) Clarity
 - (b) Enhancement
 - (c) Flexibility
 - (d) Simplicity
 - a. (a),(b) and (c)
 - b. (a),(c) and (d)** page 149
 - c. (b),(c) and (d)
 - d. All of above
4. Goto statements violate the idea of
 - a. object oriented code
 - b. structured code** page 164
 - c. control structure
 - d. repetition structure
5. A function should not be larger in any case and should not exceed _____ in length
 - a. Half page
 - b. One page** page 150
 - c. Two pages
 - d. Three Pages

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6. Data Centered Architectural Style is also called ____
- a. **Repository model** page 129
 - b. Client Server model
 - c. Sub System model
 - d. Reference model
7. Global variables in C++ should always be referred to by using the
- a. **::operator** page 153
 - b. :operator
 - c. Without an operator
 - d. None of the given
8. There are ____ layers in OSI reference model.
- a. 5
 - b. 6
 - c. **7** page 134
 - d. 8
9. Fat Client Model is one of the configurations of ____ Model.
- a. Data-centered
 - b. Layered
 - c. Reference
 - d. **Client Server** page 133
10. Dynamic process model shows the process ____ of the System.
- a. Components
 - b. Objects
 - c. **Structure** page 129
 - d. Linkage
11. Which of the following shows a commented statement in C++
- a. # Ans = first + second
 - b. **// Ans = first + second** page 162
 - c. \\ Ans = first + second
 - d. /# Ans = first + second
12. ____ architectural model is widely used in Mainframe application.
- a. Client-Server Model
 - b. **Repository Model** page 130
 - c. Filter Model

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d. Layered Model

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13. Variables should be initialized where they are _____ and they should be declared in the _____ scope possible.
- a. defined and smallest
 - b. declared and medium
 - c. defined and medium
 - d. **declared and smallest** page 161
14. Identifier names also play a significant role in enhancing the _____ of a program,
- a. Writ ability
 - b. **Readability** page 151
 - c. Reliability
 - d. All of the given choices
15. MVC stands for _____
- a. **Model View Controller** page 140
 - b. Modern View Center
 - c. Model View Center
 - d. Modern View Controller
16. Which of the following is not a possible server in client server environment?
- a. Database Server
 - b. Transaction Server
 - c. File Server
 - d. **Time Server** page 132
17. In case of using unrelated operators in a single expression, _____ would be the best choice to prevent the logical errors.
- a. Comments
 - b. Indents
 - c. **Parenthesis** page 166
 - d. Short Cuts
18. In case of header files, construction is to avoid _____ errors, the construction should appear in the top of the file (before the file header).
- a. **Compilation** page 160
 - b. run time
 - c. logical
 - d. All of the given

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19. A self documenting code is a code that explains itself without the need of comments and extraneous documentation, like__
- a. Flowcharts
 - b. UML diagrams
 - c. Process-flow state diagrams
 - d. **All of the given choices** **page 150**
20. Which one of these represents the Krutchen's 4+1 architectural view model
- a. **Logical view, Process view. Physical view. Development view. Use case view**
 - b. Logical view. Dynamic view. Physical view. Development view. Use case view
 - c. Logical view. Process view, Physical view. Development view. Sequence view
 - d. Dynamic view, Process view. Physical view, Development view. Use case view
21. Client server is a _____ system model.
- a. Integrated
 - b. **Distributed** **page 130**
 - c. Heterogeneous
 - d. Homogeneous
22. Vertical partitioning is also known as _____
- a. Balancing
 - b. Mutating
 - c. Parallelizing
 - d. **Factoring** **page 135**
23. "is" prefix should be used for _____ variables and methods.
- a. General
 - b. **Boolean** **page 152**
 - c. Constant
 - d. None of the given
24. Thin client model places a heavy processing load on _____
- a. Only server
 - b. Only network
 - c. **Both server and network** **page 133**
 - d. Neither server not network

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25. C++ header files should have the extension _____. Source files can have the extension ____.
- a. .c and .h
 - b. .cc and .h
 - c. **.h and .cpp** page 158
 - d. All of the given
26. Which of following is among ten things, which the basic template of GOF design pattern includes.
- a. IQ
 - b. **Context** page 142
 - c. Performance
 - d. Requirements
27. Vertical partitioning divides the architecture application from a _____ making perspective.
- a. **Decision** page 135
 - b. Design
 - c. Conclusion
 - d. Move
28. Issues like system performance, availability, scalability, and security are analyzed in _____.
- a. Logical View
 - b. **Physical View** page 127
 - c. Code View
 - d. Concurrency View
29. Class variables should never be declared public. Public variables violate which of the following:
- a. Information hiding
 - b. Encapsulation
 - c. **Information hiding and Encapsulation** page 161
 - d. None of the give
30. Charles Simonyi first discussed Hungarian Notation, He was of _____.
- a. **Microsoft** page 152
 - b. IBM
 - c. Dell
 - d. Cisco

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31. Patterns are devices that allow programs to share knowledge about their _____

a. Code

b. Design page 137

c. Analysis

d. Implementation

32. Process view in Krutchen's 4+1 architectural view model captures _____

a. Object model of the design

b. Concurrency and synchronization aspects of the design page 122

c. Mapping(s) of the software onto the hardware and reflects its distributed aspect

d. Static organization of the software in its development

environment 33. In the N-Tire architecture, the idea is to enhance scalability and _____ by

disturbing both the data and the application using multiple server machines

a. Usability

b. Performance page 134

c. Interoperability

d. Integrity

34. Names representing methods and functions should be _____ and written in mixed case starting with _____ case

a. Noun _____ lower

b. Verb _____ lower page 153

c. Noun _____ upper

d. Verb _____ upper

35. Floating point constant should always be written with decimal point and at least _____

a. One decimal page 164

b. Two decimal

c. Three decimal

d. none of the given

36. Static structural model shows the major system _____

a. Leaks

b. Components page 129

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- c. Activates
- d. Objectives

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37. Distributing the responsibilities to different subsystems so that we get a software system which is easy to maintain, is called_____ the architecture
- a. Subtracting
 - b. Partitioning** **page 138**
 - c. Cloning
 - d. Balancing
38. It ensures that a class only has one instance and provides a global point of access to it
- a. Singleton pattern** **page 145**
 - b. Observer pattern
 - c. Real pattern
 - d. None of the given
39. N-tier architecture stems from the struggle to find a ___ between the fat-client architecture and the thin client architecture
- a. Concurrency
 - b. Distribution point
 - c. Middle ground** **page 134**
 - d. Similarity
40. In client server model, the application is modeled as a set of ___ that are provided by servers.
- a. Request
 - b. Protocols
 - c. Service** **page 130**
 - d. Requirements
41. Maintainability and portability issues are discussed in _____
- a. Functional view
 - b. Physical view
 - c. Code view** **page 126**
 - d. Logical view
42. In case of a file server, client request selected records from a ___ and the server transmits recorded to client over the network
- a. Local memory
 - b. Network

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c. Database

d. File

page 132

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43. Physical view in Krutchen's 4+1 architectural view model captures _____.
a. Object model of the design
b. Concurrency and synchronization aspects of the design
c. **Mapping(s) of the software onto the hardware and reflects its distributed aspect**
d. Static organization of the software in its development environment
44. Logical view in Krutchen's 4+1 architectural view model captures _____.
a. Concurrency and synchronization aspects of the design
b. Mapping(s) of the software onto the hardware and reflects its distributed aspect
c. **Object model of the design**
d. Static organization of the software in its development environment
45. When programmers use shortcuts and cryptic codes in their program, _____ becomes a major problem.
a. **Maintenance**
b. Reusability
c. Availability
d. Readability
46. Which one is correct?
a. **double total = 0.5;**
b. double total = .5;
c. double total = .50;
d. all of the given
47. "Description of communicating objects and classes that are customized to solve a general design problem in a particular context" is called _____.
a. **Design Pattern**
b. System Pattern
c. System Design
d. Design Method
48. Development view in Krutchen's 4+1 architectural view model captures
a. Object model of the design
b. Concurrency and synchronization aspects of the design
c. Mapping(s) of the software onto the hardware and reflects its

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distributed aspect

- d. **Static organization of the software in its development environment**

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49. Client server model tries to _____ data and processing.

- a. **Distribute**
- b. Merge
- c. Clone
- d. Proceed

50. Which indent size eliminates the chance of code lines splitting?

- a. **2**
- b. 4
- c. 3
- d. 6

51. Which one of these is a correct short cut used by the programmer?

- a. **$x^*=a$**
- b. $x=*a$
- c. $x*a=$
- d. $=a*x$

52. _____ was the first pure Object Oriented language in which observer pattern was used in implementing its Model View Controller pattern

- a. **Smalltalk**
- b. PASCAL
- c. JAVA
- d. C++

53. Identify the TRUE statement

- a. Portability effects Security
- b. **Size of data types vary from one machine to other**
- c. Size of control structures vary from one machine to other
- d. None of the given

options 54. 1) $x = (a + 2 > 3) ? a : a - 1;$

2) $\text{if}((a + 2) > 3)$

$x = a;$

else

$x = a - 1;$

- a. **Statement (1) is more complex than (2)**
- b. Statement (2) is more complex than (1)
- c. Both statements are very complex
- d. None of the given options

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55. which of the following statements are same in

output1) $a = a \gg 2$

2) $a = a/4$

3) $a = a * 2$

a. (1) and (3) only

b. (2) and (3) only

c. (1) and (2) only page 168

d. All produce the same result

56. Code should not be:

a. **Commented** page 162

b. indented

c. **cryptic** page 164

d. aligned

57. If $!(\text{block} < \text{activeBlock})$ is

equivalent to:

a. $\text{if} (\text{block} < \text{activeBlock})$

b. $\text{if} ((\text{block} == \text{activeBlock}))$

c. **$\text{if} ((\text{block} \geq \text{activeBlock}))$** page 163

d. None of the given

58. The form for $(;;)$ should be used for

a. Nested loop

b. **Empty loop** page 159

c. More than 1000 iteration

d. Less than 1000 iterations

59. using proper parenthesis normally makes the code

a. easy to read

b. **easy to understand** page 167

c. less ambiguous

d. **All of the given option**

60. The clients and servers, in a client server environment, are connected through

_____.

a. Framework

b. Interface

c. **Middleware** page 129

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d. Groupware

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61. _____provides a unified interface to a set of intelligence in a sub-system

- a. Observer pattern
- b. Singleton pattern

c. Façade pattern

page 143

- d. All of the above

62. Complex expressions:

- a. Make the code easy to modify

b. Make the code difficult to modify

page 164

- c. Make the code easy to understand
- d. Does not effect understandability

63. Which of the following is used for multi level commenting?

- a. //

Comment **b.**

/*Comment*/

- c. (Comment)
- d. */Comment/*

1. Bring-in system's a view and define, from the system's perspective, the the software functionally the developers must build.

- **Functional Requirements**
- Non-Functional Requirements
- User Requirements
- Business Requirements

2. Which of the following is NOT an objective for building an analysis model?

- Define set of software requirements
- Describe customer requirements
- **Develop an abbreviated solution for the problem**
- Establish basis for software design

3. Consider the following scenario:

“Student logs the system and checks his/her lecture schedule”

Keeping in mind the use case diagram, and above scenario the one of the post conditions might be:

- Students should have a login
 - System should be available
 - Students should log-off after checking schedule
 - Lecture schedule should be displayed date wise
4. OOA is intended to define-----, their relationships, and their behavior.
- Variables
 - Classes
 - Objects
 - Subjects
5. One of the most powerful features of exception handling is that an error can be-----over function boundaries.
- Thrown
 - Called back
 - Caught
 - Sent
6. Inspections can check conformance with a specification but not conformance with the-----requirements.
- Customer's real
 - Developer
 - Tester
 - Manger
7. For inspections, -----are prepared that contain information regarding defects.
- Tables
 - Checklists
 - Lists
 - Farms
8. The first "bug" was actually a moth, which flew through an open window and into one of the Mark -----'s relays.
- II
 - I
 - III
 - IV
9. Holistic medicine, concerns itself with the state of the body as a whole, not the-----that is currently attacking it.
- Target
 - Reason

- Disease
 - Source
10. What factor has no precipitation in more sophisticated and complex computer-based systems?
- Vast use of personal computers
 - Vast increases in computer memory and storage capacity.
 - Greater variety of exotic input/output options.
 - Profound changes in computer architectures.
11. What types of errors are missed by black-box testing and can be uncovered by white-box testing?
- Runtime errors
 - Logic errors
 - Performance errors
 - Input errors
12. When large amount of data is to be shared, repository model is used. This model has been extensively used in the-----based application.
- Mainframe
 - Super Computers
 - Personal Computers
 - Real time
13. A(n)-----is a variance from a desired product attribute.
- Error
 - Exception
 - Defect
 - Mistake
14. -----ensures that a class only has one instance and provides a global point of access to it.
- Singleton Pattern
 - Observer Pattern
 - Behavioral Pattern
 - Joint Pattern
15. Performance & usability are example of-----requirements.
- Business
 - Functional
 - Non-Functional
 - User

16. CamelCase is now the official convention for file names and identifiers in the-----programming language.

- C#
- C++
- Java
- Visual Basic. Net

17. Comments should be indented relative to their position in the-----.

- Code
- Design
- Analysis
- Requirements

18. One of the guidelines to avoid common mistakes is to never use-----except for declaration.

- ,
- :
- =
- II

19. When planning for performance, one should always remember the-----rule.

- 80/20
- 70/20
- 60/20
- 100/20

20.

Behavioral class pattern uses-----relationship to distribute behavior between classes.

- Composition
- Aggregation
- Association
- Inheritance

21. Software crisis appeared in-----.

- Early 50s
- Early 60s
- Early 70s
- Early 80s

22. In C++ or java, named constants are written in all uppercase letters with-----to separate words.

- Semicolon;

- Underscore
- Colon:
- Dot.

23. Indentation larger than-----makes deeply nested code difficult to read and increases the chance that the lines must be split.

- 2
- 4
- 3
- 1

24. Which phase of software development lifecycle is considered most expensive?

- Requirement Gathering
- System Design
- Maintenance
- Development

25. A software requirement document describes all the-----provided by the system along with the constraints under which it must operate.

- Conditions
- Services
- Tasks
- Actions

26. -----and functional specification documents are produced in requirement analysis phase.

- Feasibility report
- Requirement definition
- Requirement specification
- Design definition

27. During the program execution most of the time is spent on-----.

- Conditional Structure
- Loops
- Arrays
- Functions

28. Many applications need to be ported on to many different-----.

- Platforms
- Language
- Compilers
- Software

29. There are-----steps involved in identification of structure while deriving object model.

- Two
- Three
- Four
- Five

30. The technique which is used to separate error-handling code from normal code is called-----.

- Exceptional handling
- Function handling
- Code handling
- Variable handling

31. In order to show the presence of a defect, a-----breaks the system.

- Tester
- Developer
- Requirement engineer
- Designer

32. Architectural model proposed by Clements et. al. contains-----views.

- 3
- 4
- 5
- 6

33. -----server provides set of applications that enable communication among clients using text, images, bulletin boards, video, etc.

- File
- Database
- Transaction
- Groupware

34. -----is result of efforts to find a middle ground between the fat client architecture and the thin-client architecture.

- Pipe and filter architecture
- N-tier architecture
- Three tier architecture
- Zero install

35. -----server provides set of applications that enable communication among clients using text, images, bulletin boards, video, etc.

- File

- Database
 - Transaction
 - Groupware
- 36.-----is result of efforts to find a middle ground between the fat client architecture and the thin-client architecture.
- Pipe and filter architecture
 - N-tier architecture
 - Three tier architecture
 - Zero install
- 37.The process view captures the concurrency and-----aspects of the design.
- Synchronization
 - ASynchronization
 - Mapping
 - Data
- 38.-----involve purely black box testing.
- Unit testing, Beta testing
 - Acceptance testing, Interfacing
 - Beta testing, Acceptance testing
 - Integration testing, Interface testing
- 39.-----is the correct formula for calculating Cyclomatic complexity of a program.
- $V - N + 2$
 - $E - V + 2$
 - $E - N + 2$
 - $E + N - 2$
- 40.-----is not included in test criteria applied in a phase of testing.
- Functional validity
 - Interface integrity
 - Correctness
 - Programming Logic
- 41.The nature of software application can be characterized by their information-
-----.
- Complexity
 - Content
 - Determinacy
 - Content and determinacy
- 42.The best way to conduct a requirements validation review is to-----.

- Examine the system model for errors
- Have the customer look over the requirements
- Send them to the design team and see if they have any concerns
- Use a checklist of questions to examine each requirements

43. Flow charts represent-----.

- Sequence Activity
- Random Activity
- Parallel Activity
- Shuffle Activity

44. A process which does not take input is called-----.

- Miracle process
- Core process
- Secondary process
- Zombie process

45. State Transition Diagram is helpful in determining-----.

- Business understanding
- Process flow
- Data store
- Control flow

46. In a-----, each program module is represented by a rectangular box.

- Use case diagram
- Document flow diagram
- Class diagram
- Data flow diagram

47. Which of the following is not a fundamental structured programming construct?

- Recursion
- Condition
- Repetition
- Sequence

48. Which writing style is best regarding identifier role in enhancing the readability of a program?

- If (Flag==0)
- If (Flag==START_NUMBER)
- If (Z==START_NUMBER)
- If (Z==0)

49.Exception handling is a powerful technique that separates error-handling code from-----code.

- Normal
- Abnormal
- Single
- Complex

50.One of the most powerful features of exception handling is that an error can be-----over function boundaries.

- Thrown
- Called
- Caught
- Sent

51.Inspections can check conformance with a specification but nit conformance with the-----requirements.

- Customer's real
- Developer
- Tester
- Manger

52.For inspections, -----are prepared that contain information regarding defects.

- Tables
- Checklists
- Lists
- Frames

53.Static analyzers are software tools for-----processing.

- Analysis text
- Source text
- Design text
- Maintenance text

54.What factor has no precipitation in more sophisticated and complex computer-based systems?

- Vast use of personal computers
- Vast increase in computer memory and storage capacity
- Greater variety of exotic input/output options
- Profound changes in computer architectures

55.Testing is an intellectually demanding activity and has a life cycle-----to software development.

- Equal
- **Parallel**
- Sequential
- In contrast

56.-----are used to describe flow of data or control in an application.

- Code structures
- String matchings
- Paths
- **Flow graphs**

57. Writing test cases and generating test data are processes that demand----- building capabilities.

- Code
- Domain
- System
- **Scenario**

58. Idea of zero install architecture is to develop a system where no installation on the-----is needed.

- **Client side**
- Server side
- Client & server side
- Network

59. There are four basic coding structures: sequence, if statement, case statement, and-----.

- For loop
- **While loop**
- Switch statement
- Logical operations

60. The pattern movement became very quiet until-----when patterns appeared again at OOPSLA conference.

- **1987**
- 1988
- 1962
- 1995

61. In-----coverage scheme, all possible paths of a program from input instruction to the output instruction are tested.

- Branch coverage
- Statement coverage

- Path coverage
 - System coverage
62. Inspections cannot check-----characteristics.
- Non-Functional
 - Business
 - User
 - Functional
63. Split lines occur when a statement exceed the-----column limit.
- 80
 - 90
 - 95
 - 85
64. Comments should be indented relative to their position in the-----.
- Code
 - Design
 - Analysis
 - Requirements
65. STL is a library of reusable components provided by-----compilers.
- Java
 - Cobol
 - Small talk
 - C++
66. In-----, the responsibility of instantiation of an object is given to some other object.
- Behavioral
 - Structural
 - Creational
 - Observer
67. -----pattern is an example of creational pattern.
- Façade
 - Singleton
 - Observer
 - Prototyping
68. -----pattern hides the implementation of the subsystem from clients, making the subsystem easier to use.
- Observer
 - Fecade

- Creational
- Singleton

69. Indentation larger than-----makes deeply nested code difficult to read and increase the chance that the lines must be split.

- 2
- 4
- 3
- 1

70.-----language allows short circuiting.

- Pascal
- C
- C++
- ADA

71. Which phase of software development lifecycle is considered most expensive?

- Requirement gathering
- System design
- Maintenance
- Development

72. Dynamic memory allocation is done from internal memory storage called---.

- Cache
- Pool
- VRam
- Boot Memory

73. Context level diagram is also known as-----.

- 0-level DFD
- 1-level DFD
- 2-level DFD
- 3-level DFD

74. Unit testing process is done by-----stakeholder of the software.

- Project manager
- Customer
- Software
- Developer

75. The number of paths in a program that contains loops tends to-----.

- Finite
- Infinity

- Limited
- Countable

76.UML is among different-----which are used for documenting the object oriented design.

- Reports
- Documents
- **Notations**
- SRS

77.White box testing is performed to test the _____ of the program a. Logic

b. Structure page 198

- c. Flow
- d. Syntax

78.Some bit field members are stored:

- i. left to right
 - ii. right to left
 - iii. in circular array
- b. only (I) is true
- c. Only (II) is true
- d. Only (III) is true

e. Both (I) and (II) are true page 183

79.In order to write a portable code which of the following guideline will not be helpful:

- a. Stick to the standards
- b. Program in the mainstream
- c. Size of data types

d. Using vendor specific language extensions page 179

80.. _____ is qualitative benefit of Unit Testing

a. Confidence building **page 207**

- b. Repeatability
- c. Bounded code
- d. Cost effectiveness

81. In order to make a code more portable. Instead of using vendor specific language extensions, use _____ as much as possible

a. STL **page 179**

- b. ANSI
- c. ISO
- d. CMM

6. In system development life cycle, defects cannot enter into the program during _____ phase.

- a. Requirement
- b. Design
- c. Coding

d. Implementation **page 209**

7. `str = 0; // str is string`

Which rewritten form of above line of code is more in line with the self-documentation philosophy than the code above.

a. `str = false;`

b. `str = NULL;` **page 169**

c. `str = \0';`

d. `str = 0.0;`

8. _____ and _____ are important short circuiting logical operators.

- a. AND b. OR c. NOTd. NOR

a. a & b pae 173

- b. b&c
c. c&d
d. a & c.

9. _____ testing isolates every part of the program and shows that the individual parts are correct.

- a. White Box
b. Black Box

c. Unit google

- d. System

10. _____ is not a white box testing technique.

- a. Statement Coverage
b. Branch Coverage
c. Path Coverage

d. State Transition Coverage page 202

11. Chip level testing for hardware is equivalent to _____

- a. White Box
b. Black Box

c. Unit testing page 207

- d. System testing

12. For equivalence partitions, we divide the problem in _____ obvious categories.

- a. Two**
- b. Four
- c. Five
- d. Six

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13. `x = 0; // x is floating pt`

Which rewritten form of above line of code is more in line with the self-documentation philosophy than the code above?

- a. `x = false`
- b. `x = NULL`
- c. **`X = 0.0;`**
- d. `x = '\0'`

page 169

14. One of the main reasons to make functions is _____

a. Reliability

b. Reusability

page 171

- c. Maintainability
- d. Efficiency

15. 80/20 rule states that:

a. you spend 80 percent of your time in 20 percent of the code

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- b. you spend 20 percent of your time in 80 percent of the code
- c. We should try to optimized 80 percent or at least 20 percent of the code
- d. We should try to optimized 20 percent or at least 80 percent of the code

16. _____ identifies which statements in a method or class have been executed.

a. Method Coverage

b. Statement Coverage

- c. Class Coverage
- d. Branch Coverage

17. In _____ system's internal implementation details are not visible to the tester.

a. White Box Testing

b. Gray Box Testing

c. Black Box Testing **page 198**

d. Both White Box Testing & Gray Box Testing

18. Verification and validation are the processes in which we check a product against its _____ and the _____ of the users who will be using it

a. statements, expectations

b. specifications, expectations **page 192**

c. specifications arguments

d. statements, arguments

19. Objective of _____ is to show that the program does not work

a. Requirements

b. Design

c. coding

d. testing **page 196**

20. The formula for calculating the Cyclomatic Complexity of a program is:

a. $E - N + 2$ **page 208**

b. $E + N + 2$

c. $E - N - 2$

d. $N - E + 2$

21. The raising of the imaginary error flag is simply called raising or _____ an error

a. Catching

b. Casting

c. Throwing page 184

d. Repress

22. Be very careful when you use functions with side effects - functions that change the values of the _____

a. Objects

b. Classes

c. Structures

d. Variables

Be very careful when you use functions with side effects – functions that change the values of the parameters. Page 176

23. The number _____ is the most abused symbol in programs written in C or C++.

a. 1

b. 0 page 169

c. 2

d. 3

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24. _____ is a creative activity while _____ is a destructive activity

a. Development, Testing **page 196**

- b. Testing, Development
- c. Validation, Verification
- d. Verification, Validation

25. Which one is used to describe flow of data or control in an application

- a. Class diagram
- b. Use case diagram

c. Flow Chart **page 200**

- d. Code

26. Testing individual components independent of other components is called:

a. Unit testing **page 189**

- b. Module testing
- c. Subsystem testing
- d. System testing

27. The expressions with logical operators can be evaluated only from _____

- a. Right to left

b. Left to right **page 173**

- c. Top to bottom
- d. Bottom to top

28. The complexity of a program may _____ if there are exceptional paths in it

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a. Decrease

b. Increase page 185

c. Remain same

d. Cutback

29. In the switch statement, cases should always end with a _____ statement.

a. Switch

b. Go

c. Break page 167

d. Stop

30. Cyclomatic complexity measures the logical complexity_____

a. Quantitatively page 205

b. Qualitatively

c. Creditably

d. Worthily

31. The size of _____ plays a significant role in making the program
easy or difficult to understand

a. Function page 170

b. object

c. data type

d. none of the above

32. _____ depicts programming instructions that do not have branching

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or any control information

a. Sequence **page 200**

- b. IF
- c. While
- d. Case

33. for (i =0, col = 0; i < 27; i++.j++)

In the above line of the code 27 is representing _____.

- a. Real Numbers

b. Magic Numbers **page 167**

- c. Constant Numbers
- d. Positive Numbers

34. The process that involves verification of product with respect to its written requirements is called _____

- a. Maintenance
- b. Debugging
- c. Development

d. Testing **page 192**

35. If an application fulfills its specifications but deviates from users expectations or their desired behavior. This means, software is verified but not -----

a. Validated **page 192**

- b. Corrected
- c. Checked
- d. Traced

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36. A _____ variance from a desired product attribute.

- a. Exception
- b. Error
- c. Mistake

d. Defect **page 192**

37. If a function changes the value of some other accessible data object along with returning its value after execution is called _____

- a. Short Circuiting
- b. Modularity

c. Side Effects **page 174**

- d. Abstraction

38. if (e.Title() == "CEO" || e.Salary() > 10000)

in the above statement if (e.Title()=="CEO") is TRUE then

a. Second part is not evaluated **page 185**

- b. Second part is always evaluated
- c. Second part must also be TRUE
- d. Second part is not evaluated but e.Salary ()is called

39. When an error is thrown the overall system (on the lookout for this error flag) responds by _____ the error

- a. Ignoring
- b. Casting

c. Catching **page 184**

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d. Pass over

40. Testing of collection of modules to discover interfacing problems among interacting modules is called

a. Unit testing

b. Module testing

c. Subsystem testing page 198

d. System testing

41. Constant values used in the program are called a. Real Numbers

b. Magic Numbers page 168

c. Constant Numbers

d. Positive Numbers

42. Bit fields are a convenient way to express many difficult operations. However, bit fields suffer from one problem

a. Lack of usability

b. Lack of security

c. Lack of performance

d. Lack of portability page 183

43. Comma (,) is very dangerous because _____

a. Compiler does not recognize this symbol

b. It creates linkage problem

c. It causes side effects page 176

d. It does not causes side effects

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44. Function testing falls under _____ testing technique

- a. White Box
- b. Black Box
- c. Unit**
- d. System

page 207

45. The greatest advantage of exception handling is its ability to handle:

- a. Asynchronous errors**
- b. Syntax errors
- c. Memory errors
- d. Control Structure errors

page 184

46. After integrating subsystems into a system, and then testing this system as a whole is called

- a. Unit testing
- b. Component testing
- c. Subsystem testing

d. System testing

47. struct packed_struct{

unsigned int f1:1;

} pack;

Here in "packed_struct":

a. value of f1 = 1

b. size of f1 = 1 bit

page 183

c. value of f1 should not exceed 1

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d. None of given options

48. The C/C++ language has not specified whether _____ is arithmetic or logical.

a. Right shift >> **page 181**

b. Right shift <<

c. &&

d. ||

49. A _____ is a variance from a desired product attribute

a. Exception

b. Error

c. Mistake

d. Defect **page 192**

50. There are _____ basic coding structures.

a. One

b. Two

c. Three

d. Four **page 200**

51. In _____ control can take either of several branches.

a. IF

b. While

c. Case statement **page 201**

d. Sequence

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52. _____ cause major portability issues

- a. Loops
- b. Bugs in code

c. Sizes of data types **page 179**

- d. Conditional Structures

53. A good program must contain _____ infeasible paths.

a. 0 **page 206**

- b. 1
- c. N
- d. Infinite

54. We do not use flow graphs to _____.

- a. Show flow of data

b. Describe decisions **page 200**

- c. Show Control in an application
- d. Describe different coding structures

55. The order in which bytes of one word are stored is _____ dependent

a. Hardware **page 181**

- b. software
- c. language
- d. syntax

56. _____ is the starting point, base document for both testing and the development

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- a. Software code
- b. Test case document
- c. Software architecture document

d. Functional specification document page196

57. if a major rework is required to translate a program written for one environment to another, it means code is not or less _____

- a. Secure
- b. Bug Free
- c. platform dependent

d. Portable page 179

58. Which of the following is the correct path for the code:

If(a%b==0) 1

C=a+b; 2

Cout<<c; 3

- a. 1-2-3-1
- b. 1-3-2
- c. 1-2-3**
- d. 2-3-1

59. Switch statement is equal to _____ statement.

- a. if
- b. if-else

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c. nested if

d. if-else if page 167

60. Exception handling provides

- a. Portability mechanism
- b. Code Usability mechanism

c. Error Handling mechanism page 184

d. Both Portability & Code Usability mechanism

61. Unit Testing will be done by

- a. Testers
- b. End Users
- c. Customer

d. Developers page 207

62. Identify the TRUE statement

- a. Portability effects security

b. Size of data types vary from one machine to other page 179

- c. Size of control structures vary from one machine to other
- d. To write portable code, use vendor specific language extensions as much as possible

63. Which type of testing is not concerned with how the inputs are transformed into outputs?

- a. White Box

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b. Gray Box

c. Black Box page 198

d. Both White Box & Gray Box

64. Consider the following statement: `int a,b=10;`

Which of the following is correct: a. variable "a" is initialized to 10

b. Variable "b" is initialized to 10

c. Both variables "a" and "b" are initialized to 10

d. variables can not be initialized this way

65. Software _____ is the process of examining the software product against its requirements.

a. Testing page 192

b. Debugging

c. Exception handling

d. Fixing

66. The idea behind exception handling is to raise some error flag every time

a. The code compiles

b. The code links

c. Memory is allocated

d. Something goes wrong page 184

67. Bit fields allow the packing of data in a structure, using Bit fields we can:

a. Read 9 bit integers page 182

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- b. Avoid memory leakages
- c. Avoid memory overflow
- d. Avoid syntax errors

68. _____ is a tool that can help us in reducing the size of individual functions.

- a. Inheritance

b. Modularity page 170

- c. Association
- d. Abstraction

69. _____ and _____ are two important tools that helps in managing the program complexity

- a. Composition, Inheritance

b. Abstraction, Encapsulation page 170

- c. Aggregation, Inheritance
- d. Modularity, Composition

70. A path through a program which is never traversed for any input data is called

- a. Dependent path
- b. Independent path

c. Infeasible path page 206

- d. Feasible path

71. When a small set of functions (which use each other) is so overwhelmingly the bottleneck, there are two alternatives:

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a. use a better algorithm OR re-write the code **page 177**

b. debug the code OR place assertions in code

c. remove the functions OR add more functions

d. changed programming language OR compiler at least

72. Alpha testing is:

a. Testing individual components independent of other components

b. Testing a collection of dependent components

c. Validation against user expectations

d. Acceptance testing for customized projects, in-house testing for

products

page 198

73. Exception handling is a powerful technique that separates error-handling code from _____ code.

a. Normal **page 184**

b. Faulty

c. Buggy

d. Complex